

igaFund:

Educational funding platform for verified underprivileged youth talents in Sub-Saharan Africa.

- **Description of the concept of their Mission**

igaFund platform is a web based educational funding platform designed to connect verified underprivileged African talented youth with donors, scholarship providers and institutions across sub saharan Africa. My mission is to reduce educational inequality by creating a transparent, secure and accessible platform where talented but financially disadvantaged students can receive verified educational support through me, you in crowdfunding or also organisations as well as scholarship providers.

Subsaharan African continues to face educational access crises especially in rural and low income communities where students fail or drop out of school due to lack of school fees and limited financial resources. Existing crowdfunding platforms are not designed for these realities as they generally contribute money to personal accounts or wallets and the purpose of the donation isn't surely confirmed it was used for the same purpose.

igaFund addresses the gap by introducing verified student profiles, ambassador programs for help in reaching wider communities, administrative verification workflows and direct to institution payment routing. This will create transparency, donor trust and assurance but overall contributing to increased rate of educational access for African youth.

- **Problem statement**

Who is affected?

The people who are affected by this problem are students in Sub-Saharan Africa. These students are mostly school graduates and university applicants from rural areas and low-income communities. Donors and scholarship providers are also having a time finding trusted ways to fund education.

What issue do they have?

The main issue here is that there is no way for disadvantaged African students to get funding for their education. The existing crowdfunding systems are not working well in areas with internet connectivity. They also do not have mechanisms in place to verify the students and ensure that the funds are being used properly.

When do they encounter the issue?

This problem is always there, it gets worse when students are moving from lower secondary to upper secondary education and from secondary school to university. At these times students have to pay a lot of fees including registration fees, tuition, uniforms and examination costs.

Where is the issue happening, which communities face this the most?

This problem is happening over Sub-Saharan Africa. However the initial plan is to start solving this problem in Rwanda. Rwanda has a digital infrastructure and a well-developed mobile money ecosystem making it a good place to start.

Why are many struggling with the issue?

Many young people from families cannot continue their education because they do not have enough money. They also do not have access to funding for education. There are no trusted systems in place to help them. Donors are also hesitant to give money because they cannot verify if the students are genuine or if their money is being used properly.

How are we solving this problem?

igaFund is going to solve this problem by creating a website that can be used by everyone. This website will have profiles of students that have been verified. It will be easy for donors to contribute money. The website will also make sure that the money is being used properly and it will send notifications to donors and students by email and SMS. The website will work well in areas, with poor internet connectivity making it easy for everyone to use including igaFund and the students it is helping igaFund and its team.

- **Software development model**

Agile Scrum software development model

The Agile scrum software development model is the framework selected for the igaFund platform. This is because the platform does not have fixed requirements from development and planning. It has user workflows such as student registration, ambassador offline data capture, admin verification, donor browsing, and payment routing which will be shaped significantly by feedback from real users during pilot testing. A Waterfall model would require all requirements to be finalized before a single line of code is written, which is inappropriate for a product being built in a context we are still learning. Agile Scrum allows the team to build, test, learn, and adapt in iterations without delaying the entire system whenever a user needs changes. Additionally this is going to be developed in a modular manner of systems with clearly separable functional units: the student web portal, the ambassador mobile application, the admin verification dashboard, the SMS gateway integration, and the payment routing system.

The below Agile Scrum's 8 week sprint structure maps directly onto this modularity where each sprint delivers one testable, deployable module rather than attempting to build everything simultaneously.

Sprint 1 (Weeks 1–2) User Registration and Authentication Module: Building student, ambassador, donor and administrator registration workflows, email and password authentication with bcrypt hashing, role-based access control and password recovery via email. Sprint review with test user accounts to validate registration flow usability and functionality.

Sprint 2 (Weeks 3–4) Student Profile Management and Ambassador Onboarding: Develop student profile creation, support document uploads, guardian consent form capture, ambassador-assisted student onboarding online or offline with local encrypted storage and automatic synchronization when internet connectivity is restored. Sprint review with testing pilot ambassador accounts to test reliability.

Sprint 3 (Weeks 5–6) Admin Verification Dashboard and Donor Contribution Portal: Building administrative profile review for approval or rejection workflows, users account management, guardian consent verification for minors, audit logging, donor browsing of verified profiles, contribution processing and direct-to-institution payment routing via future integrated gateways. Sprint review with facilitators and test donor accounts to validate simulated payment routing accuracy.

Sprint 4 (Weeks 7–8) Notification System, Analytics, Testing and Deployment: Delivery of email notification integration for profile status changes, funding milestones and payment confirmations, funding analytics dashboard and report export functionality. Conduct full system integration testing, cross-browser compatibility testing, security audit and deploy igaFund V1.0 MVP to cloud infrastructure. Final sprint to document lessons learned, get user feedback for V1.0 pilot and plan V2.0 backlog priorities and features.

- **The hypothesis of their solution**

We anticipate that by deploying igaFund platform which is a verified student profile platform combining self-service online registration with community ambassador assisted enrolment, internal administrative profile verification, guardian consent documentation, direct-to-institution payment routing. The platform is expected to achieve the following measurable outcomes within the first 12 months of operation after deployment:

- Create and activate 50 verified student profiles in Rwanda within 6 months of launch, each linked to a confirmed educational institution with a validated payment routing information.
- At least 60% of activated profiles receive partial or full tuition funding within 90 days of going live. Success is confirmed by an institution-issued payment receipt.
- Rwanda's existing mobile money transactional systems such as MTN MoMo and Airtel Money, REB digital exam records and developed smartphone penetration in the country makes the technical foundation viable for a small pilot program to run and test its acceptance with real users.
- Rwanda's tertiary transition rate declined to 49.6% (MoE, 2025) and an annual financing gap of US\$70 billion regionally (UNESCO, 2023) confirm both the demand for the platform and the scale of available capital to utilize for the success of the platform and impact making as intended.
- Most features of the igaFund V1.0 in MVP terms shall be delivered within this development lifecycle that includes but not limited to student portal, ambassador interface, admin dashboard, notification gateway methods, and payment routing

information. After successful launch of v1.0 and collecting user feedback the igaFund platform will be developed to V2.0 to then add in institutional scholarship marketplace for partnerships and collaborations.

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1. Introduction

1.1 Purpose

This software requirements document helps define the functional and non-functional requirements for igaFund version 1.0. This is a web based education funding platform made for underprivileged youth in sub-Saharan Africa but with an initial pilot market being Rwanda. This document is going to demonstrate technicality referencing and evaluating the design, development and testing of the platform. We will be covering the requirements for igaFund V1.0 MVP release then V2.0 which will then have an institutional scholarship marketplace.

This software requirements specification covers requirements for igaFund V1.0 Minimum Viable Product (MVP) which includes student registration or onboarding, ambassador onboarding, student profile verification, donor funding through crowdfunding or group funding, notifications via email or SMS, direct institution payments routing. The future V2.0 release will be introduced after partnering with institutions for providing scholarship opportunities to verified profiles that are then prioritized for those opportunities, in this document we will explicitly specify V1.0 but not much scope about V2.0.

1.2 Document Conventions

The following standards and typographical conventions are being used in this software requirements specifications (SRS) document:

- Calibri is the font we are using, text size 11 aligned and indented, bolded text is used for emphasis and showing keywords and for titles.
- Section heading follows numerical numbering structure to improve organization.
- Functional requirements are identified using prefix FR followed by numerical numbering format.
- Non functional requirements are identified using prefix NFR followed by numerical numbering.
- Business rules are identified using the prefix BR.
- User roles are Students, Donor, Ambassador and system administrator are clarified and consistent.
- All requirements are written clearly and in testable language to support validation and traceability throughout the software development life cycle.

1.3 Intended Audience and Reading Suggestions

This document is intended for multiple stakeholders involved in the development and growth of igaFunda project, including:

- **Software developers** who reference this documentation for designing, building and implementing the system.
- **ALU course facilitators** assess academic quality, research, project validness and completeness of the project documentation.
- **Quality assurance and testing teams** responsible for validating the system functionalities and performance metrics.
- **Project management** that oversees the planning, execution and delivery of the product.
- **Future institutional partners and investors** who evaluate the platform's capabilities and impactful potential to the African community.

The document is organised into major sections that cover the system description, requirements, business rules, references and appendices. Sections include:

1. **Introduction** which is to help understand the project background, development and objectives.
2. **Description** that helps to understand the system context and user expected environment.
3. **Requirements specifications** which is to review detailed functional and non functions requirements of the system.
4. **Appendices** will help support definitions, glossary items and models analysis.

1.4 Product Scope

igaFund is an educational funding platform developed to reduce financial constraints and information barriers that prevent talented underprivileged African youth from accessing secondary and tertiary education opportunities.

The platform combines education technology (EdTech) and financial technology (FinTech) to connect verified student profiles directly with donors through crowdfunding, scholarship providers, educational institutions through a secure and transparent platform. Key objectives of the platform are to:

- Reduce dropout rates in secondary education for financially struggling students.
- Increase retention to tertiary education to a higher rate for talented disadvantaged students.
- Improve donor trust through verified student records and secure payment routing by eliminating misuse of donated funds for other activities but directing the payment architecture to institutions directly.
- Create a scalable education funding ecosystem across sub Saharan Africa.
- Support rural offline communities using ambassador-assisted registrations.

The igaFund V1.0 release focuses on:

- Student self registration.

- Ambassador onboarding and assisted registration for students (Offline or online).
- Administrative control and verification workflows
- Donor(s) contribution management.
- Notifications systems via SMS or email
- Secure direct to school institutions payment architecture and routing.

2. Overall Description

2.1 Product Perspective

The igaFund platform is a new standalone educational financing platform designed to address financial and accessibility barriers affecting underprivileged young talented African youth seeking education and opportunities to grow their talents. The platform is designed specifically for the African context with an initial pilot implementation focus in Rwanda. It's not to replace or directly integrate with existing platforms or tools but it is designed to fill a gap left by general purpose fundraising or crowdfunding platforms where usage of the funds is not assured of the purpose it was contributed for.

igaFund functions as a verified educational funding ecosystem that connects students to donors and educational institutions through a secure digital platform by introducing routing of direct payments to institutions ensuring that donated funds are used strictly for educational purposes. The platform is designed as a three-tier architecture consisting of:

- A frontend web interface for students, ambassadors, donors and administrators
- A backend application layer responsible for logic, verification and payment processing routing and workflows.
- A relational database management system used for secure storage of student and donor records, transactions and audit logs.
- In addition to the web platform, there is also a mobile application support and responsiveness to support users who use the platform by their mobile smartphones with also limited internet connectivity.

The system will also interface with different external service providers for communications, payment processing and verification via Application Program Interfaces (APIs) including:

- MTN Mobile Money API
- Airtel Money API
- VISA/Bank card payment gateways
- SMS gateway APIs

2.2 Product Functions

igaFund platform V1.0 provides a secure and verified educational funding ecosystem that enables students,donors,ambassadors and educational institutions to interact or connect through a centralized digital platform. The major functions of the system include:

- **Student management functions:** Individual student profile or ambassador assisted registration, document upload, educational background and funding information then progress tracking.
- **Verification and administrative functions:** Administrative review and verification of submitted student profile data, approval and rejection workflows, guardian consent verification, role based access control for system users.
- **Donor funding functions:** Public student profiles discovery and browsing, donor account registration, contribution processing, direct to institution payment routing, receipts upload and funding confirmation proofs.
- **Communication functions:** Automated notification for profile updates,funding statuses or progress, guardian and student notification, payment to donors via emails or SMS
- **Mobile and offline support functions:** Offline student profile capture, local encrypted storage of unsynchronized records to automatically synchronize for internet connectivity availability and mobile responsive access for smartphone users.

2.3 User Classes and Characteristics

The igaFund platform will be used by multiple categories of users with different levels of responsibilities, functions and system privileges. The most important user classes for the success of the platform are students, ambassadors, administrators and donors as they directly participate in all workflows of the funding process.

User role	Role description	System access
Student	Creates and owns their own student profile containing personal background, academic records, supporting documents, information, funding targets, and individual goals.	Can monitor finding progress, receive updates, no modification of their profiles after approval.
igaFund Community ambassador	A current or alumni who is a beneficiary of the platform responsible for assisting in onboarding qualified rural and underserved talented youth in their communities or areas of reach.	Register a student profile, capture information, upload documents, submit student profiles for reviews and monitor their submitted student profile statuses be it the platform approval or rejection or even funding goals updates.

		Notification systems for different actions.
Donor	Browsing verified and approved student profiles and contributing to their education funds goal. This can be a group of people, an individual or an organisation chooses to sponsor for the whole financial goal of a profile.	View student profiles, submit transaction receipts for submission to schools, receive updates on profiles or reminders based on their preferences.
igaFund Administrator	Review submitted student profiles, verify support documents and guardian consent forms.	Approve or reject applications, manage user roles, monitor logs and transactions, fully access system records.

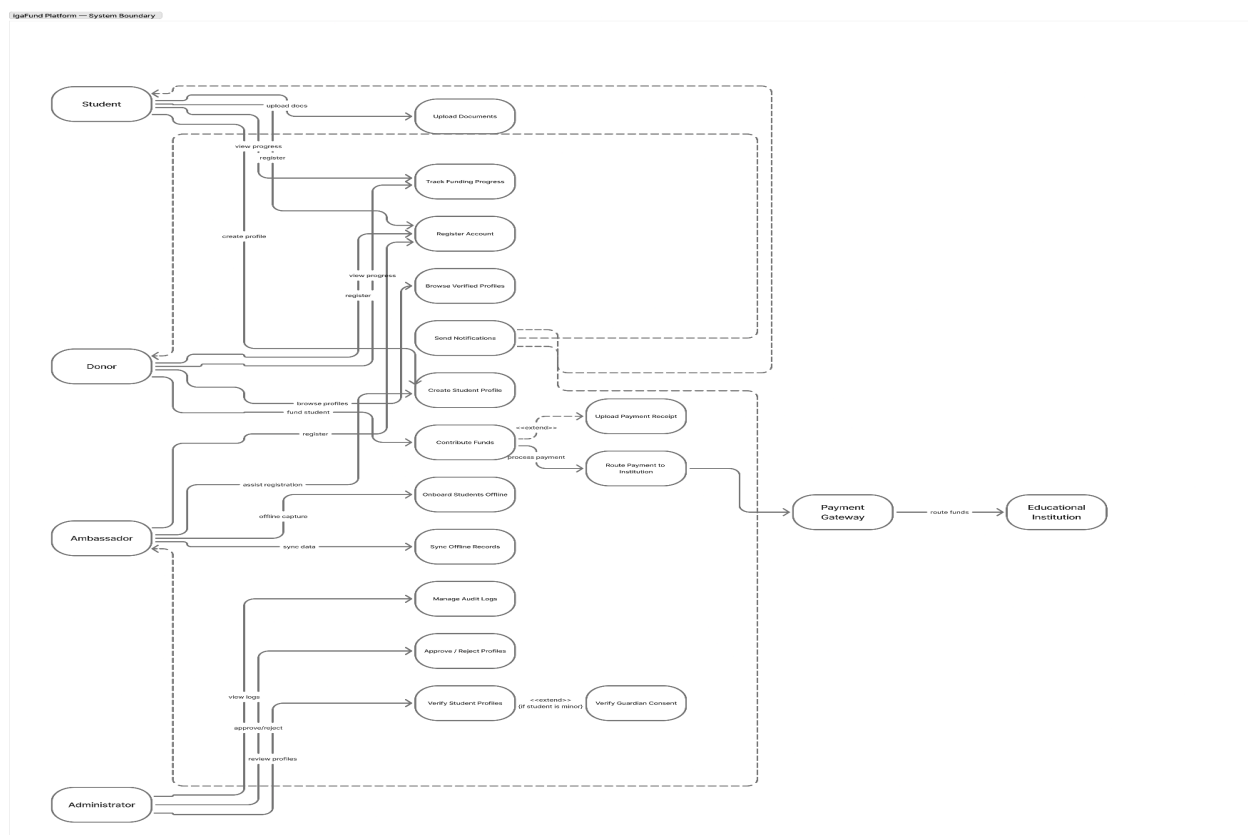


Figure 1: The use case diagram illustrates the functions of the igaFund platform by mapping the interactions between different system users.

2.4 Operating Environment

igaFund platform shall operate within the following software, hardware and network environments:

- The web application shall be accessible through standard desktop and mobile web browsers such as Google Chrome, Brave browser, Microsoft Edge and Safari.
- The frontend interface shall support mobile smartphones compatibility.
- igaFund mobile application will have offline tracking and synchronization capabilities once connection to internet is available
- Backend services shall be deployed on secure cloud infrastructure platforms with regional support to minimize latency.
- The relational database management system shall support secure storage, backup and management of data used across the platform including transactions made, student and donor records, uploaded documents and audit logs.
- Notification communications shall be integrated with SMS gateway APIs and emails for proper communications
- Payment processing will be integrated with different external service API providers for routing to direct school payment links for the funds donated.

2.5 Design and Implementation Constraints

The igaFund platform development and implementation shall be subject to the following constraints and limitations:

Regulatory and legal constraints: Where all personal information and records must comply with Rwanda's Data Protection and Privacy Law. Students who are minors must have a verified guardian or parent consent information documentation before publication on the platform.

Financial constraints: Financial disbursement shall only be routed directly to verified educational institution accounts not to personal wallets or accounts and it will be restricted by system architecture. Secure encryption mechanisms will be applied to sensitive student information, uploaded documents and transactions records.

Connectivity constraints: The platform will support offline data collection for those in under inconsistent internet connectivity in rural areas by ambassador-assisted or individually then synchronization processes must function reliably when internet connectivity becomes available.

Technology and resource constraints: The platform will depend on operational availability of external services via APIs such as SMS gateways, payment gateways, MTN Mobile Money and Airtel Money services. This will be supported for future scalability and expansion beyond the piloting area to other African regions. Initial resources deployment limit V1.0 pilot to selected pilot area which is Rwanda at the beginning and budget limitations may affect future pace of feature development and regional expansions.

2.6 User Documentation

User Manuals: Student user manuals explaining profile registration, document uploads and fundraising progress tracking. Donor user guides describing contribution workflows, how it works and transaction confirmation processes. Onboarding manuals for offline student profile registration and synchronization operations.

Online help and tutorials: Context sensitive online help within the platform interface using step by step guidance and walkthroughs for common platform activities.

Knowledge base and support: A centralized knowledge base containing troubleshooting guides and instructional content with support resources for different issues such as payments, synchronization failures and profile verification processes.

Accessibility and language support: Documentations should follow accessibility standards to support users with different disabilities and future platform expansion will have multilingual documentation support including English, French and Kinyarwanda.

Documentation delivery formats: Documents may be delivered through PDF documents, embedded in platform support systems, web based support pages and mobile accessible documentation interfaces.

2.7 Assumptions and Dependencies

Internet and technology availability

- Users are assumed to have least occasional internet connections for synchronization so communications and funding transactions are synced and corrected correctly and fully.
- Users are assumed to possess mobile smartphones capable of running the mobile web based application.

Third-party service availability

- External service provider APIs are assumed to remain operational and accessible through the V1.0 pilot phase. Communication gateway providers like emails or SMS are assumed to support reliable notification messaging delivery.

Educational institution cooperations and partnerships

- Schools partnerships are assumptions of cooperation with different segmented or fully disbursed tuition verification and verification of payment processing to keep account of a student who is being funded.

- Academic documents such as Rwanda Education Board/NESA results are assumed to exist in digital, scanned and notified by notaries to be submitted for verification processes.

User participation and adoption

- Students are required to provide accurate personal and academic information during registration which will go to an extra layer of administrative verification for confirmation. Donors will be required to responsibly comply with platform policies for data protection of the profiles and themselves.

Financial resources availability

- Availability of funding and technical resources support are assumed to be available for development, maintenance and support of the platform to get the V1.0 pilot implemented in Rwanda.

Regulations stability

- The educational and finance regulations environment in Rwanda is assumed to remain stable and supportive for educational funding initiatives and financing services.

3. External Interface Requirements

3.1 User Interfaces

The igaFund platform provides responsive and user-friendly web and mobile interfaces for students, ambassadors, donors and administrators for better experience and easy navigation.

The web platform shall support:

- Student registration and profile management
- Donor browsing and funding workflows
- Administrative dashboard interface, verification and management workflows of system users.

The igaFund platform shall also support mobile responsiveness for the mobile application:

- Offline support for community ambassadors in student onboarding.
- Mobile documents uploads support
- Synchronization tracking for accurate updated information.

The platform shall follow responsive web design principles with standards in navigation layouts, validation notifications, authentication controls and accessibility support across desktop and mobile devices.

3.2 Hardware Interfaces

The igaFund platform shall support interactions with:

- Desktop and laptop computers
- Mobile smartphones and tablets
- Smartphone cameras for document capture and scanning
- Cloud based backend servers and database for storage management systems

The platform shall communicate through secure internet protocols and REST Application services. The mobile application shall support local storage and synchronization functionality for areas with unstable internet connectivity.

3.3 Software Interfaces

The igaFund platform is built to reduce educational funding barriers for underprivileged talented African youth, it will interface with several software systems, databases and APIs from external providers. All these are necessary to support the assurance of an outcome intended for the platform's mission, supporting communications, profile management, records, payments routing and verification workflows.

Database interfaces: The platform will use a relational database management system responsible for storage and management of student, donor and ambassador records, uploaded documents, transaction records and receipts and also audit logs tracking information.

Operating system compatibility: The platform shall operate on multiple operating system environments including windows operating systems, android and apple mobile devices, linux based operating systems and macOS environments.

Third-part APIs and services; Integration of different services from third-party service providers will include SMS and payment service gateways such as MTN Mobile Money, Airtel Money and VISA/bank account gateways. These are to support communications, payment routing and transactional confirmations.

Mobile application interfaces: Mobile application will be specifically designed for ambassadors for easily onboarding profiles and it will support offline student onboarding, document uploads, status tracking and notification but this will have a synchronization process to run when internet is available.

Security and encryption interfaces: Protection of user information and financial transactions is a must and this will be implemented through secure authentication systems, encrypted communication

channels, role based access control mechanisms, password hashing and session management and an always monitoring of platform activities through logs.

Reporting and analytics interfaces: Reporting and analytics tools and dashboards to provide insights will be built to show funding progress, total funds through the platform, student verifications statistics, operations performance, administrative and financial reports.

3.4 Communications Interfaces

The platform shall support communication, synchronization and secure transaction notifications and statuses that will be implemented through several interfaces connecting users, external services and platform components.

Notifications and messaging communication interfaces: Platform will support email communication and communication protocols for reliable message delivery for account verification, funding confirmations, administrative notifications, password recovery and student donor status updates. Through SMS notifications and email communication channels. Message formats are clear and easy to understand for timely and reliable delivery.

Web browser communication: igaFund platforms shall be accessible through modern web browsers such as google chrome, safari, microsoft edge and other platforms. It should remain responsive and functional across desktop and mobile environments.

Data management and performance: The ambassador mobile platform shall have synchronization mechanisms that allow offline data collection, local encrypted storage, automatic synchronization when internet connectivity becomes available and low bandwidth optimizations for mobile low-bandwidth network accessibility.

User notifications and alerts: Communication interfaces shall support automated alerts and notifications related to profile verification statuses, funding milestones or achievements, payment confirmations and statuses, synchronization updates and administrative communications. This will all follow standard communication protocols to ensure compatibility with external APIs, cloud services, payment gateways and notification systems.

4. Requirement Specification

STAKEHOLDER REQUIREMENTS SPECIFICATION

Functional Requirements

Req ID	Requirements	Description
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FR 1	User registration and authentication	
FR 1.1	Student registration	Allow create student profiles, submit academic information, upload documents and define funding goals.
FR 1.2	Ambassador registration	Allow registration, verify identity and onboard students from their communities.
FR 1.3	Donor registration	Donors are allowed to create accounts to contribute and track donations.
FR 1.4	Administrator account	Allow administrators to manage the whole system, user accounts, permissions and CRUD operations.
FR 1.5	User login and access control	Allow all users to securely log into the platforms and be redirected based on their credentials for what they can access.
FR 1.6	Password recovery	Allow users to reset forgotten passwords using email or SMS verification processes.
FR 2	Student profile management	
FR 2.1	Student profile creation	Allow students and ambassadors to capture personal details, academic and personal background, guardian and individual information, funding needs and goals.
FR 2.2	Supporting documents upload	Allow upload of academic records, identification documents, recommendation and supporting documents and guardian consent forms if minor.
FR 2.3	Student profile edits	These will be allowed to be done before administrative verification is in progress or over a request to edit that information
FR 2.4	Funding progress tracking	Students will be allowed to track donations, payment status and funding milestones.
FR 2.5	Profile status notifications	Students, and ambassadors shall be notified when profiles are approved, rejected, update or funding status changes.
FR 2.6	Student dashboard	Students are able to view funding progress, notifications, profile status and information submitted, request for profile information adjustments or changes and payment of funds updates.

FR 3	Verification and administrative management	
FR 3.1	Administrative review	Administrators review submitted student profiles and uploaded documents for approval or rejection
FR 3.2	Guardian consent form verification	Administrators verify consent documents for minor or underaged students.
FR 3.3	Role based access control	Restrict access to features and records based on assigned user roles.
FR 3.4	Administrator dashboard	Dashboards for user management, analytics, verification workflows, reports and audit logs.
FR 3.4	Audit Logs and storage management	Records and logs are tracked and controlled by the administrator this includes records kept, transaction activities and profile updates.
FR 4	Donor contribution management	
FR 4.1	Browsing verified student profiles	Allow donors search and view verified student funding profiles.
FR 4.2	Direct institution payment contribution processing	Allow donors to contribute funds using supported payment gateways directly to education institutions accounts.
FR 4.3	Payment receipt upload	Donors upload payment confirmations and receipts for tracking in institution accounts.
FR 4.4	Donation tracking	Donors are able to monitor contribution history, funding progress and payment confirmations,
FR 4.5	Donor dashboard	Dashboards displaying verified profiles that need funding, respective donor funded profiles, funding goal milestones and progress of profiles, donation history and contribution summaries and notifications.
FR 5	Communication notification system	
FR 5.1	Email notifications	Sent for registrations verification, donations, status changes and password recovery.
FR 5.2	System alerts	Users are notified of pending actions, profile updates, funding milestones and platform updates.
FR 6	Mobile and offline support	
FR 6.1	Offline student onboarding	Ambassadors can capture student data without internet connections.

FR 6.2	Local encrypted storage	Store offline records securely on mobile before synchronization
FR 6.3	Automatic synchronization	Synchronize locally stored records once internet connectivity is restored.
FR 7	Analytics and reporting	
FR 7.1	Funding analytics	Generation of insights on donations, funding and financial statistics on the platform.
FR 7.2	Student verification statistics	Statistics to show approvals, rejections and pending profiles.
FR 7.3	Reports generation for export	Authorised users shall be allowed to export reports of different metrics in supported formats such as PDF.

5. Other Nonfunctional Requirements

5.1 Non-Functional Requirements

Requirement Type	Req ID	Description
Security	NFR 1	The system shall implement secure user authentication with Role based access control (RBAC) for all platform users. Donor and student accounts shall require email and password authentication meeting a minimum of 8 characters including uppercase letter, number, and special character. Each role and account may only access the functions and data explicitly assigned to it. User passwords shall never be stored in plain text and shall be hashed using bcrypt (minimum cost factor 12) before database storage.
Security	NFR 2	All data in transit between users, external APIs, and platform servers shall be encrypted using HTTPS be it student records, uploaded documents, guardian information, and financial transaction records shall be encrypted. No file or student record shall be accessible via a permanent public URL.
Privacy & Compliance	NFR 3	The platform shall comply with Rwanda's Data Protection and Privacy Law for all personally identifiable information, with particular enforcement for minors. No minor's legal name, photograph, precise location, or national ID number shall be visible to unauthenticated or unverified users under any system state.

Requirement Type	Req ID	Description
Audability	NFR 4	The system shall maintain complete, immutable, and timestamped audit logs for all authentication events (successful and failed), financial transaction attempts, profile status changes, document access events, and administrative actions. Each log entry shall record: user account ID, action type, affected record ID, IP address, and timestamp. These reports can be exported in PDF formats by administrators in case of need.
Introperability	NFR 5	The system shall operate and integrate with externa service providers including payment gateways and SMS or email services.
Cross Browser Support	NFR 6	The system must be accessible across web browsers both those that have mobile, desktop and tablet versions, those include Google Chrome, Microsoft Edge, Safari and others
Technology	NFR 7	The system must be accessible different devices such as laptops, desktops, android and iOS devices. And these different screens have responsiveness and mobile support for easy navigation.

5.2 Performance Requirements

Response Time

User operations such as login, dashboard loading, donor profile browsing and form submission shall respond within normal load conditions. API endpoints returning student profile data shall respond quickly as these are set to ensure usability on low-end Android smartphones common among Rwandan users.

Scalability

The system architecture shall be designed to support authenticated users without degradation in response time or data integrity. The backend shall support scaling fro growth through cloud infrastructure so that increased user growth during peak academic intake periods does not require system redesign or emergency intervention.

Reliability

The platform shall minimise downtime and maintain stable operations for all funding transactions and profile management activities. Financial transaction processing shall be either fully completed or fully rolled back to prevent partial fund routing errors.

Synchronisation Performance

Offline student profile records captured by Ambassadors shall synchronise automatically and completely once internet connectivity becomes available. The synchronisation engine shall handle queue processing sequentially to prevent record data duplication, corruption or loss.

Availability

The platform shall maintain a target uptime of 99.5 percent measured monthly. Planned maintenance windows and planned database backups shall be retained to keep the users engaged with the system always and no downtime in peak hours or when users are using the platform.

5.3 Safety Requirements

Data protection

Student records, donor information, guardian contact details, and all uploaded documents shall be stored in encrypted, access-controlled storage to prevent unauthorised access or data loss.

Secure transactions

All payment routing and financial disbursement activities shall use encrypted and authenticated communication channels. Under no circumstances shall any portion of a donor contribution be routed to a personal bank account or personal mobile money wallet.

Minor protection

Student profiles belonging to minors (under 18 years of age) shall require a verified, scanned guardian or parental consent form before any profile can be approved and published on the platform.

Administrative verification

No student profile shall become publicly accessible without passing through the internal administrative review and approval workflow. Profiles submitted through either the self-service web portal or the ambassador mobile app shall remain in a pending verification state completely hidden from the donor portal and all search indexes until an authenticated administrator explicitly approves the profile.

5.4 Security Requirements

Authentication and Authorization

The platform shall implement secure login authentication for all user roles. Accounts shall require email and password authentication meeting a minimum complexity of 8 characters including uppercase letters, numbers, and special characters. Role based access control (RBAC) shall enforce strict access boundaries each user role may only access the system functions and data explicitly assigned to that role.

Password protection

User passwords shall never be stored in plain text. All passwords shall be encrypted using a strong password hashing algorithm (bcrypt with a minimum cost factor of 12 or equivalent) before storage in the database.

Secure communications

All data transmitted between users, external APIs, and the platform servers shall use HTTPS with TLS 1.2 or higher. HTTP connections shall be automatically redirected to HTTPS.

Audit logging

The system shall maintain complete, immutable, and timestamped logs of all authentication events (successful and failed), financial transaction attempts, profile status changes, document access events, and administrative actions. Logs shall record the user account ID, action type, affected record ID, IP address, and timestamp for every event.

Session security

The system shall automatically terminate inactive authenticated user sessions after 24 hours of inactivity. Users shall be required to re-authenticate after session expiry. Concurrent session limits shall be enforced for Administrator accounts.

Compliance

The platform shall comply with Rwanda's Data Protection and Privacy Law for all user data handling, storage, and processing activities mostly for minors on the platform.

5.5 Software Quality Attributes**Usability**

The platform shall provide intuitive interfaces and simple workflows accessible to users with different technical skill levels. The student self-registration workflow shall be completable in under 10 minutes by a first-time user. All interfaces shall meet accessibility standards and be responsive across different screen widths.

Reliability

The platform shall maintain stable operations with a target uptime of 99.5 percent. All financial transactions shall be either fully completed or fully rolled back with zero partial transaction states. Ambassador offline sync shall guarantee zero data loss across all sync cycles.

Scalability

The system shall support future expansion to larger user populations and additional African countries without major architectural redesign. The cloud backend shall support horizontal scaling. The database schema shall be designed to accommodate additional country codes, currencies, and language settings without structural changes.

Maintainability

The software architecture shall be modular, with clearly separated functional components (student management, verification, payments, notifications) to support efficient maintenance and independent feature upgrades.

Interoperability

The system shall integrate effectively with MTN Mobile Money, Airtel Money, VISA/bank card payment gateways, SMS APIs and encrypted cloud document storage through well-defined, authenticated REST

API connections and shall use documented, versioned API interfaces to enable future provider changes without system redesign.

Performance Efficiency

The platform shall optimize synchronisation, document uploads, and dashboard performance for low-bandwidth environments. Profile images shall be auto-compressed. API queries shall return results within standard load timing.

Security

The platform shall protect all sensitive information through AES-256 encryption, RBAC access control, and audit logging.

Testability

All system components student management, verification workflow, payment routing, SMS notifications, and admin dashboard shall be independently testable through unit and integration tests.

5.6 Business Rules

BR 1

Only verified student profiles that have passed internal administrative review shall be published in the public donor pool and made accessible for donor browsing and funding.

BR 2

All donor contributions shall be routed directly and exclusively to the verified bank account or mobile money wallet of the student's registered educational institution. Under no circumstances shall any funds be routed to a personal account or wallet.

BR 3

Students who are minors (under 18 years of age) must have a signed, scanned guardian or parental consent document uploaded and confirmed as present in their profile record before any approval and publishing of their profile.

BR 4

Students shall not modify their profile data, uploaded documents, or funding targets once their profile has been approved by an administrator. Any updates to an approved profile shall require the profile to request with reasons and it shifts back to pending status for a new administrative review cycle.

BR 5

Community ambassadors shall only view, manage, and submit student profiles that they personally enrolled through their ambassador account.

BR 6

Donors shall only access publicly approved student profile information presented on the donor browsing portal. Donors shall not access confidential student records, full legal names, home addresses, national ID numbers, or guardian contact details unless explicitly unlocked through a verified KYC process in V2.0.

BR 7

Administrators shall have authority to approve, reject, flag, suspend, or archive student profiles and

platform records. All administrative actions must be accompanied by a mandatory written note and are permanently recorded in the system audit log.

BR 8

All users accounts and roles shall comply with platform privacy, verification, and funding policies as stated in the platform terms of service and compliance of the legal law.

BR 9

The platform shall maintain complete, immutable audit records for all financial transactions and profile verification activities. These records shall be accessible to Administrators for monitoring, compliance reporting, and accountability purposes and shall be retained according to the retention schedules defined in NFR 3 and NFR 12.

BR 10

All users shall submit accurate and verifiable information during account registration and profile submissions. The platform employs an additional administrative verification layer to confirm submitted information before any profile is made public. Submission of false or misleading information shall result in immediate account suspension.

6. Appendix

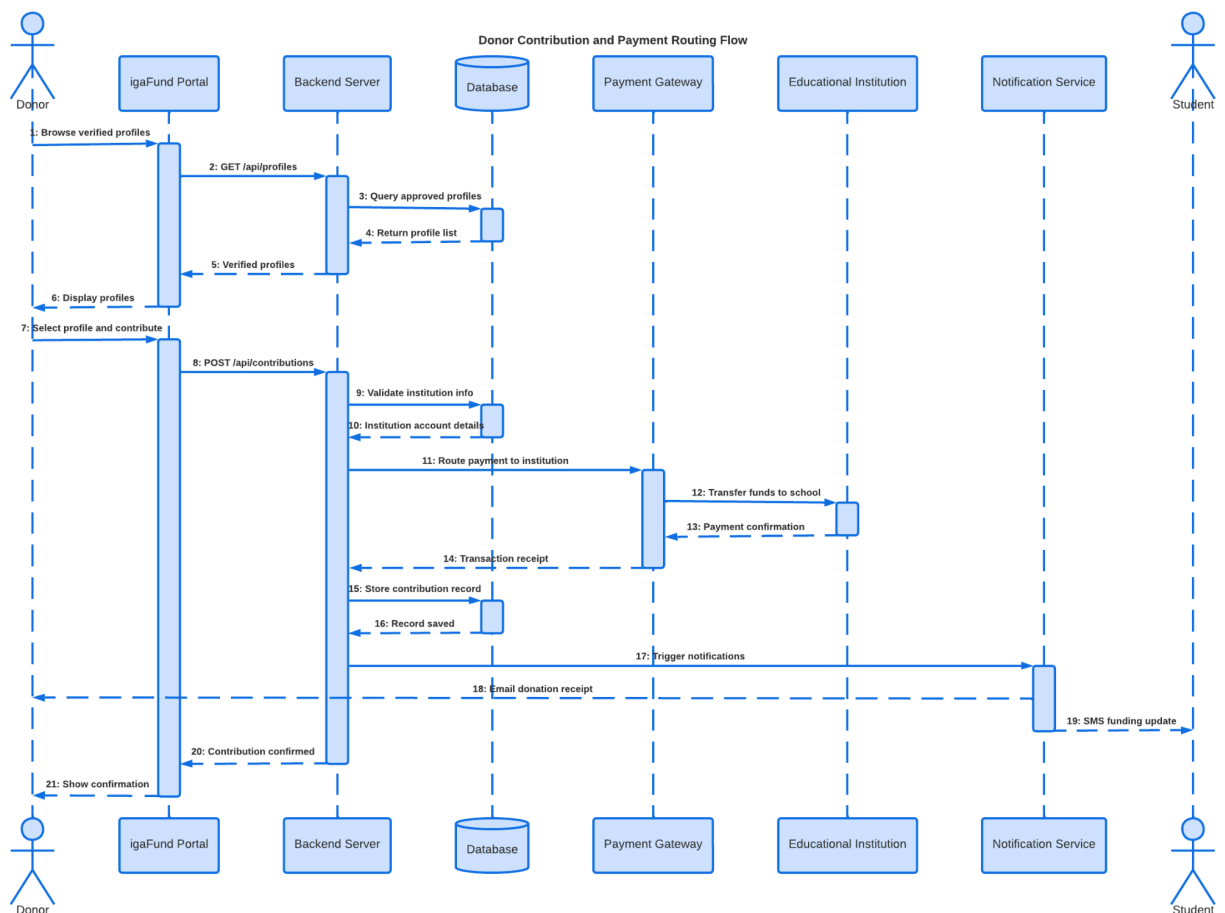
Appendix A: Glossary

- igaFund: the term “iga” is a kinyarwanda word for learning meaning learning Funds.
- API: Application Programming Interface used for communication between external systems and the igaFund platform.
- Ambassador: A verified community representative responsible for assisting students with registration and onboarding, especially in underserved communities who benefited or are currently benefiting from the platform.
- Audit Log: A permanent record of user and system activities used for monitoring, accountability and security tracking.
- Crowdfunding: The process of collecting educational financial support from multiple donors for a student profile.
- Donor: An individual, organisation, or group contributing funds toward verified student educational needs.
- EdTech: Educational Technology systems and digital tools supporting education services.
- FinTech: Financial Technology systems used for secure digital financial transactions and payment processing.
- HTTPS: HyperText Transfer Protocol Secure used for encrypted communication between users and the platform.
- KYC: Know Your Customer verification process used to confirm user identity and prevent fraud.
- Local Storage: Secure temporary storage of offline records on a mobile device before synchronization.
- MVP: Minimum Viable Product representing the first functional release of the igaFund platform.
- NESAI: National Examination and School Inspection Authority in Rwanda responsible for academic examination records.

- Offline synchronization: Process of uploading locally stored records to the cloud server once internet connectivity becomes available.
- RBAC: Role Based Access Control mechanism restricting system access according to assigned user roles.
- REST API: Web service architecture used for communication between software systems and external providers.
- SMS Gateway: External service provider responsible for delivering SMS notifications and alerts.
- Student Profile: A verified digital record containing student academic, personal, and funding information.
- Verification workflow: Administrative review process used to approve or reject submitted student profiles and documents.
- igaFund platform V1.0: Initial pilot release focused on verified educational crowdfunding workflows.
- igaFund platform V2.0: Planned future release introducing scholarship marketplaces and institutional partnership features.

Appendix B: Analysis models

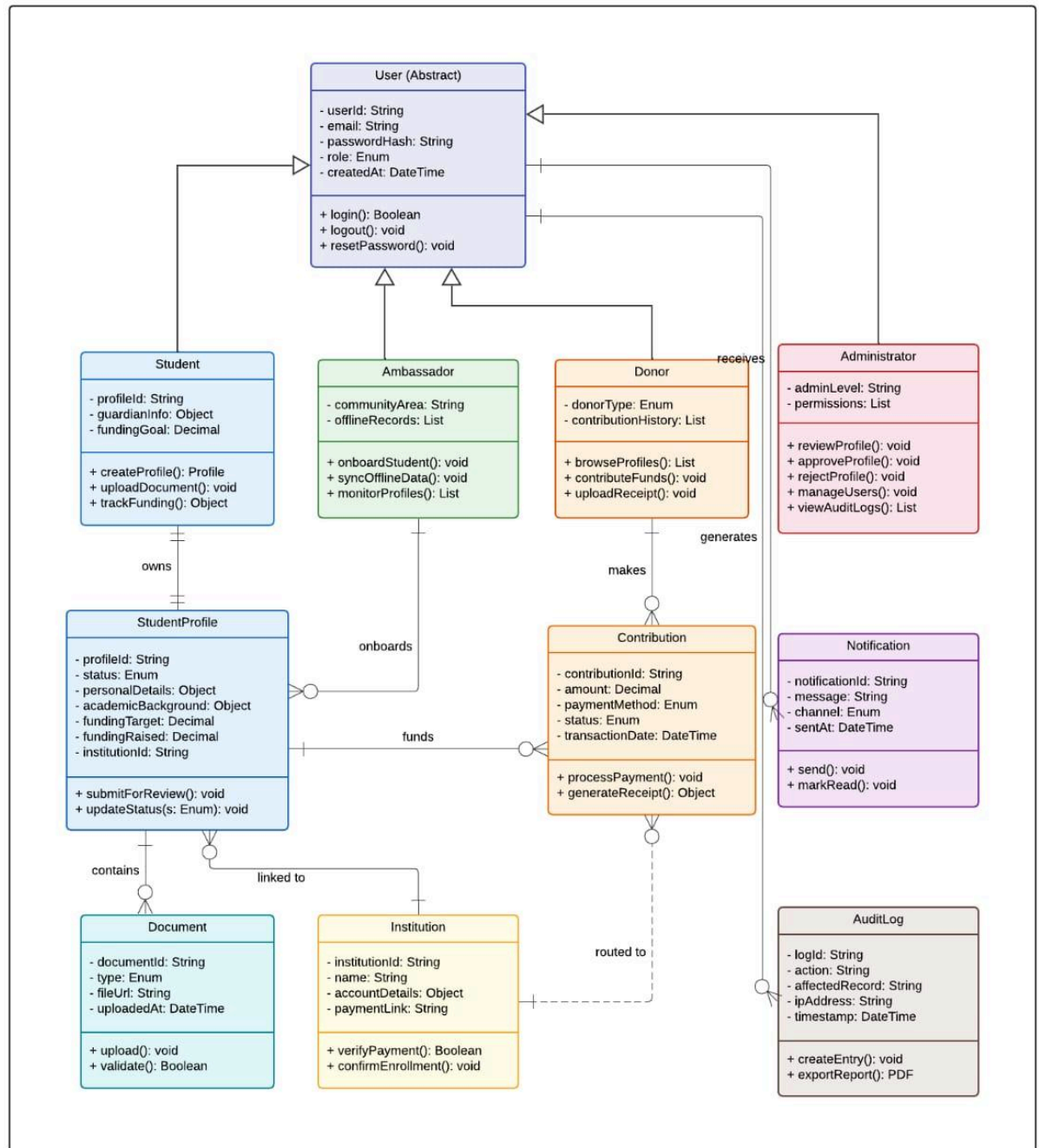
1. Sequence Diagram:



This sequence diagram models the Donor Contributions and Payment Routing flow, the

platform's core workflow. Eight participants take part: the Donor, igaFund Portal, Backend Server, Database, Payment Gateway, Education Institution, Notification Service and Student. It runs in three phrases: browsing verified profiles, processing the contribution and routing payment straight to the institution and delivering email and SMS confirmations. Activation bars show each component processing in turn, and dashed return arrows show the system waiting for each step to finish before moving on.

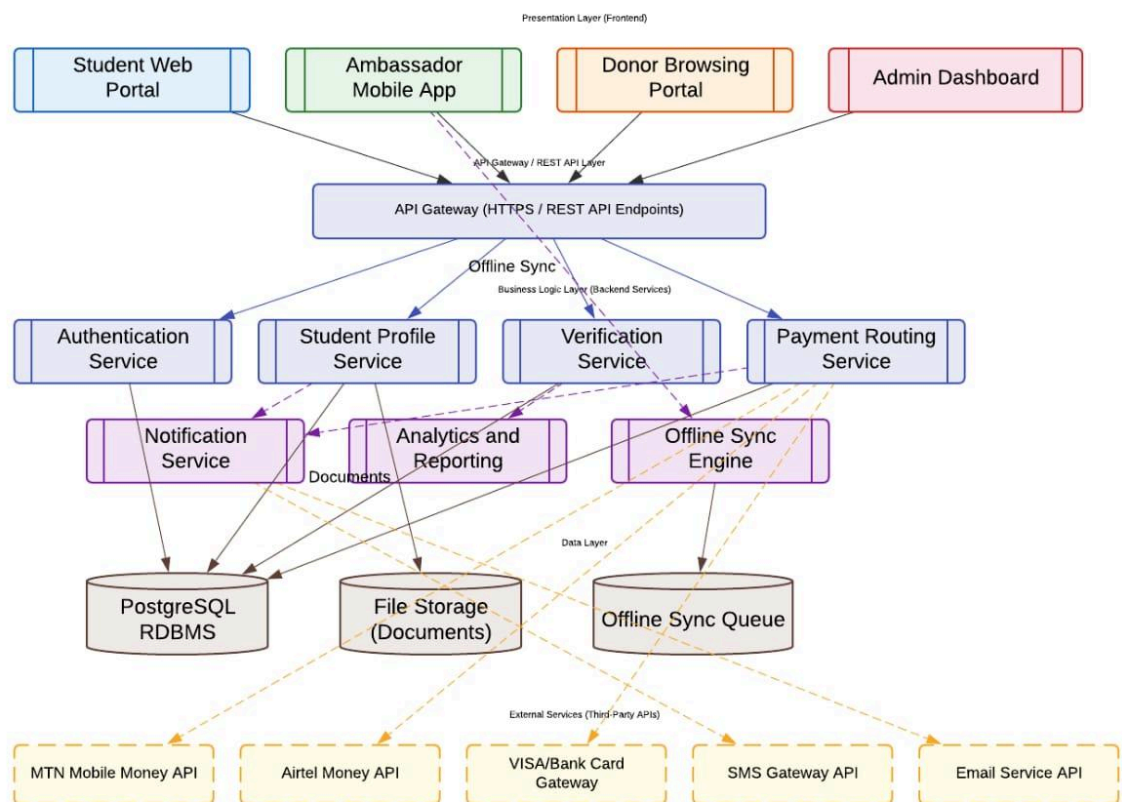
2. Class diagram:



This Class diagram shows the static structure of the igaFund platform across eleven classes, User is the parent holding shared login data and methods: Student, Ambassador, Donor and Administrator inherit from it, each adding role specific behavior.

The operational: classes Student Profile, Contribution, Document, Institution, Notification and AuditLog store the platform's funding and verification data. Relationships include inheritance, associations with multiplicity where a student owns one StudentProfile, a Donore makes many contributions and a dependency where a contribution is routed directly to an institution, enforcing BR2 that funds never reach a personal account.

3. Components diagram:



This component diagram shows the platform's layered architecture in five tiers. The Presentation layer holds the four role based frontends such as Student Portal, Ambassador App, Donor Portal, and Admin dashboard. They all reach the backend through the API Gateway, which enforces HTTPS and role based access. The Business Logic Layer holds the core services like Authentication, Student Profile, Verification, Payment Routing, Notifications, Analytics, and the Offline Sync Engine. These connect to the Data Layer (PostgreSQL, File Storage, Offline Sync) Queue and the External Services like MTN, Airtel, Visa payment APIs, SMS and Email gateways,

shown with dashed borders as thirds-party dependencies. Solid arrows are internal dependencies where the dashed ones are external.

3. Use case diagram



This use case diagram shows interactions between seven actors and almost fifteen use cases inside the platform. Four primary actors are student, ambassador, donor and administrator each positioned outside the boundary and connected by solid association lines to the use cases they initiate. Three external system actors shown with dashed borders at the bottom are the Payment Gateway, Educational Institution and SMS/Email Gateway representing third-party API integrations.

Three include relationships (blue dashed arrows) enforce mandatory prerequisites: create student profile includes upload documents meaning documents must be submitted before review, Contribute Funds includes Route Payment to Institution enforcing BR 2 that all funds go directly to school accounts never personal wallets, and Review Student Profile includes Verify Guardian Consent ensuring BR 3 compliance for minors. Three extend relationships (red dashed

arrows) model conditional behaviors: Capture Data Offline extends Create Student Profile activating only when ambassadors lack internet connectivity, and Send Notifications extends both Approve/Reject Profile and Contribute Funds triggering email and SMS alerts only after those actions complete successfully.